MKD

2026-06-24

Contest (1)

dbg.h

```cpp
#define dbg(...) cerr<<"DEBUG:["<<#__VA_ARGS__<<"] = [",\
    _print(__VA_ARGS__),cerr<<"]\n"
template<class T, class=void>struct cntr : false_type {};
template<class T>struct cntr<T,
    void_t<decltype(declval<T>().begin()),
    decltype(declval<T>().end()),
    typename T::value_type> > : true_type {};
template<class T>using raw=remove_cv_t<remove_reference_t<T> >;
template<class T>enable_if_t<!cntr<T>::value>
    _print(const T &x) { cerr << x; }
template<class T>enable_if_t<cntr<T>::value>_print(const T &v){
    cerr << "["; bool f = false;
    for (const auto &x: v)
      f ? cerr << (cntr<raw<decltype(x)> >::value ?",\n":", "):
        cerr, f = true, _print(x);
    cerr << "]";
}
template<class T, class... A>
void _print(const T &x, const A &... a) {
    _print(x);((cerr << ", ", _print(a)), ...);}
```

template.cpp

```cpp
#include <bits/stdc++.h>
using namespace std;
// #define LOCAL_DEBUG
#ifdef LOCAL_DEBUG
#include "dbg.h"
#else
#define dbg(...) 42
#endif
#define rep(i, a, b) for(int i = a; i < (b); ++i)
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
using ll = long long;
using pii = pair<int, int>;
using vi = vector<int>;
void init() {}
void solve() {}
signed main() {
    init();
    cin.tie(nullptr), ios::sync_with_stdio(false);
    // cin.exceptions(cin.failbit);
    int T = 1;
    // cin >> T;
    while (T--)solve();
    return 0;
}
```

1.1 checker

cek.sh

```sh
#!/usr/bin/env bash
set -u
cd "$(dirname "$0")"
g++ gen.cpp -o gen -std=c++17 -O2
g++ std.cpp -o std -std=c++17 -O2
g++ sol.cpp -o sol -std=c++17 -O2
cnt=0
while true; do
  cnt=$((cnt + 1))
  echo "Testing case #$cnt"
  ./gen "${1-}" > in.txt
  ./std < in.txt > out1.txt
  ./sol < in.txt > out2.txt
  if ! diff -q out1.txt out2.txt > /dev/null; then
    echo "WA on case #$cnt"
    diff out1.txt out2.txt
    read -r -p "Press Enter to continue..."
  fi
done
```

cek.bat

```bat
@echo off
cd "%~dp0"
g++ gen.cpp -o gen -std=c++17 -O2
g++ std.cpp -o std -std=c++17 -O2
g++ sol.cpp -o sol -std=c++17 -O2
set cnt=0
:loop
set /a cnt+=1
echo Testing case #%cnt%
gen %1 > in.txt
std < in.txt > out1.txt
sol < in.txt > out2.txt
fc out1.txt out2.txt > nul
if errorlevel 1 (
    echo WA on case #%cnt%
    fc out1.txt out2.txt
    pause
)
goto loop
```

gen.h

```cpp
uint64_t seed = chrono::steady_clock::now()
  .time_since_epoch().count() ^ random_device{}();
mt19937 rng(seed);
template<class T>
T randi(T l,T r){return uniform_int_distribution<T>(l,r)(rng);}
vi randvi(int n, int l, int r) {
  vi v(n);
  rep(i, 0, sz(v))v[i] = randi(l, r);
  return v;
}
vi randp(int n) {
  vi p(n);
  iota(all(p), 1);
  shuffle(all(p), rng);
  return p;
}
int main(int argc, char *argv[]) {
  if (argc >= 2)
    seed = stoull(argv[1]), rng.seed(seed);
  cerr << seed << endl;
  return 0;
}
```

Data structures (2)

UnionFind.h

Description: 并查集数据结构。
Time: $\mathcal{O}(\alpha(N))$

7aa27c, 14 lines

```cpp
struct UF {
  vi e;
  UF(int n) : e(n, -1) {}
  bool sameSet(int a, int b) { return find(a) == find(b); }
  int size(int x) { return -e[find(x)]; }
  int find(int x) { return e[x] < 0 ? x : e[x] = find(e[x]); }
  bool join(int a, int b) {
    a = find(a), b = find(b);
    if (a == b) return false;
    if (e[a] > e[b]) swap(a, b);
    e[a] += e[b]; e[b] = a;
    return true;
  }
};
```

RMQ.h

Description: 区间最小值查询。返回 min(V[a], V[a + 1], ... V[b - 1])。
Usage: RMQ rmq(values);
rmq.query(inclusive, exclusive);
Time: $\mathcal{O}(|V| \log |V| + Q)$

510c32, 15 lines

```cpp
template<class T>struct RMQ {
  vector<vector<T>> jmp;
  RMQ(const vector<T>& V) : jmp(1, V) { // pw = 2^k-1
    for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
      jmp.emplace_back(sz(V) - pw * 2 + 1);
      rep(j,0,sz(jmp[k]))
        jmp[k][j] = min(jmp[k - 1][j], jmp[k - 1][j + pw]);
    }
  }
  T query(int a, int b) {
    assert(a < b); // or return inf if a == b
    int dep = 31 - __builtin_clz(b - a);
    return min(jmp[dep][a], jmp[dep][b - (1 << dep)]);
  }
};
```

FenwickTree.h

Description: 树状数组。计算前缀和 a[0] + a[1] + ... + a[pos - 1]，并通过计算新旧值之差更新单个元素 a[i]。
Time: Both operations are $\mathcal{O}(\log N)$.

219f0b, 21 lines

```cpp
struct FT {
  vector<ll> s;
  FT(int n) : s(n) {}
  void update(int pos, ll dif) { // a[pos] += dif
    for (; pos < sz(s); pos |= pos + 1) s[pos] += dif;
  }
  ll query(int pos) { // sum of values in [0, pos)
    ll res = 0;
    for (; pos > 0; pos &= pos - 1) res += s[pos-1];
    return res;
  }
  int lower_bound(ll sum) {// min pos st sum of [0, pos] >= sum
    // Returns n if no sum is >= sum, or -1 if empty sum is.
    if (sum <= 0) return -1;
    int pos = 0;
    for (int pw = 1 << 25; pw; pw >>= 1)
      if (pos + pw <= sz(s) && s[pos + pw - 1] < sum)
        pos += pw, sum -= s[pos-1];
    return pos;
  }
};
```

Number theory (3)

3.1 整除性

欧几里得

$$\gcd(a,b)=1$$
$$\gcd(a,b)=\gcd(b,a \bmod b)$$
找符合$ax+by=d=\gcd(a,b)$的一组x,y
$$a\cdot a_{\text{inv}}+py=1=\gcd(a,p)$$
$$by+(a \bmod b)x=d$$
$$ax+b(y-\left\lfloor\frac{a}{b}\right\rfloor x)=d$$

euclid.h
Description: 找到两个整数 x 和 y，使得 $ax+by=\gcd(a,b)$。如果你只需要 gcd，可以使用内置的 __gcd。如果 a 和 b 互质，那么 x 就是 a (mod b) 的逆元。
Time: $\mathcal{O}(\log\min(a,b))$ 33ba8f, 5 lines

```
ll euclid(ll a, ll b, ll &x, ll &y) {
  if (!b) return x = 1, y = 0, a;
  ll d = euclid(b, a % b, y, x);
  return y -= a/b * x, d;
}
```

中国剩余定理

定理 1 (CRT)**.** 设 $m_1,m_2,\ldots,m_n$ 两两互质，且
$$(S):\begin{cases}x\equiv a_1 \pmod{m_1},\\ x\equiv a_2 \pmod{m_2},\\ \quad\vdots\\ x\equiv a_n \pmod{m_n}.\end{cases}$$
令
$$M=\prod_{i=1}^{n}m_i,\qquad M_i=\frac{M}{m_i},\qquad t_i\equiv M_i^{-1} \pmod{m_i}.$$
则方程组 (S) 有解，并且一组解为
$$x\equiv\sum_{i=1}^{n}a_it_iM_i \pmod{M}.$$

构造性证明. 由于 $m_1,m_2,\ldots,m_n$ 两两互质，对任意 i，有
$$\gcd(M_i,m_i)=1.$$
因此存在 t_i，使得
$$t_iM_i\equiv 1 \pmod{m_i}.$$
令
$$x=\sum_{k=1}^{n}a_kt_kM_k.$$
固定任意 i。当 $k\neq i$ 时，由于 $m_i\mid M_k$，有
$$a_kt_kM_k\equiv 0 \pmod{m_i}.$$
于是
$$\begin{aligned}x&=a_it_iM_i+\sum_{k\neq i}a_kt_kM_k\\&\equiv a_i\cdot 1+\sum_{k\neq i}0 \pmod{m_i}\\&\equiv a_i \pmod{m_i}.\end{aligned}$$
因此 x 同时满足所有同余式，故 x 是方程组 (S) 的一组解。 □

CRT.h
Description: 中国剩余定理。
crt(a, m, b, n) 计算 x，使得 $x\equiv a \pmod m$，$x\equiv b \pmod n$。如果 $|a|<m$ 且 $|b|<n$，则 x 将满足 $0\le x<\text{lcm}(m,n)$。假设 $mn<2^{62}$。
Time: $\log(n)$
"euclid.h" 04d93a, 7 lines

```
ll crt(ll a, ll m, ll b, ll n) {
  if (n > m) swap(a, b), swap(m, n);
  ll x, y, g = euclid(m, n, x, y);
  assert((a - b) % g == 0); // else no solution
  x = (b - a) % n * x % n / g * m + a;
  return x < 0 ? x + m*n/g : x;
}
```

3.2 模运算

ModularArithmetic.h
Description: 用于模运算，设置 MOD 后使用。
"euclid.h" 2f9ed0, 18 lines

```
constexpr ll MOD = 17; // change to something else
struct Mod {
  ll x;
  Mod(ll _x) : x((_x % MOD + MOD) % MOD) { }
  Mod operator+(Mod b) { return Mod((x + b.x)%MOD+MOD)%MOD); }
  Mod operator-(Mod b) { return Mod((x - b.x)%MOD+MOD)%MOD); }
  Mod operator*(Mod b) { return Mod((x * b.x %MOD+MOD)%MOD); }
  Mod operator/(Mod b) { return *this * invert(b); }
  Mod invert(Mod a) {
    ll x, y, g = euclid(a.x, MOD, x, y);
    assert(g == 1);return Mod((x % MOD + MOD) % MOD);
  }
  Mod operator^(ll e) {
    if (!e) return Mod(1);
    Mod r = *this ^ (e / 2); r = r * r;
    return e & 1 ? *this * r : r;
  }
};
```

模逆元

$$\begin{aligned}&\frac{a}{b}\equiv a\cdot b_{\text{inv}}\equiv x \pmod p\\ \Rightarrow&a\equiv bx \pmod p\\ \Rightarrow&a\cdot b_{\text{inv}}\cdot b\equiv bx \pmod p\\ \Rightarrow&a\cdot(b_{\text{inv}}\cdot b-1)\equiv 0 \pmod p\\ \Rightarrow&b_{\text{inv}}\cdot b\equiv 1 \pmod p\end{aligned}$$
$$\text{inv存在}\Leftrightarrow\gcd(b,p)=1$$
$$b\cdot b_{\text{inv}}=k\cdot p+1,\ g=\gcd(b,p)\Rightarrow g\cdot b_1\cdot b_{\text{inv}}=k\cdot g\cdot p_1+1$$
$$\Rightarrow g\cdot(b_1\cdot b_{\text{inv}}-k\cdot p_1)=1$$
$$\text{数域为}\mathbb{Z}\Rightarrow g=1,\ b_1\cdot b_{\text{inv}}-k\cdot p_1=1$$

ModInverse.h
Description: 预计算模逆元，假设 LIM $\le$ mod 且 mod 是质数。 17b468, 4 lines

```
constexpr ll MOD = 1000000007, LIM = 200000;
vector<int> inv(LIM + 1, 0);inv[1] = 1;
for (int i = 2; i <= LIM; i++)
  inv[i] = MOD - (MOD / i) * inv[MOD % i] % MOD;
```

Geometry (4)

4.1 Geometric primitives

CW ::= ClockWise 顺时针

CCW ::= CounterClockWise 逆时针

Point.h
Description: 平面点类。T 为 double 或 long long 等类型（避免 int）。 de820b, 30 lines

```
constexpr double eps = 0;
template<class T> int sgn(T x) { return (x > eps)-(x < -eps); }
template<class T> struct Point {
  using P = Point;
  T x, y;
  explicit Point(T x = 0, T y = 0) : x(x), y(y) { }
  bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
  bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
  bool approxEq(P p) const { return !(sgn(x-p.x)|sgn(y-p.y)); }
  P operator+(P p) const { return P(x + p.x, y + p.y); }
  P operator-(P p) const { return P(x - p.x, y - p.y); }
  P operator*(T d) const { return P(x * d, y * d); }
  P operator/(T d) const { return P(x / d, y / d); }
  T dot(P p) const { return x * p.x + y * p.y; } // 点乘
  T cross(P p) const { return x * p.y - y * p.x; } // 叉乘
  // Oa与Ob叉乘
  T cross(P a, P b) const { return (a-*this).cross(b-*this); }
  T dist2() const { return x * x + y * y; } // 到原点距离平方
  double dist()const{return sqrt((double)dist2());}//到原点距离
  // 与x轴夹角，范围[-pi,pi]
  double angle() const { return atan2(y, x); }
  P unit() const { return *this/dist(); } // 单位化
  P perp() const { return P(-y, x); } // 绕原点逆时针旋转九十度
  P normal() const { return perp().unit(); } // 法向量
  // 绕原点逆时针旋转a弧度
  P rotate(double a) const {
    return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
  friend ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.x << "," << p.y << ")"; }
};
```

$$dis=\frac{|(e-s)\times(p-s)|}{|e-s|}$$
$$dis\le eps\Leftrightarrow|(e-s)\times(p-s)|\le|e-s|\cdot eps$$

sideOf.h
Description: 返回点 p 相对于线段 se 的位置。1/0/-1 $\Leftrightarrow$ 左侧/在线上/右侧。如果给定了可选参数 eps，则当点 p 距离线段 se 的距离小于 eps 时，返回 0。P 应该是 Point<T> 类型，其中 T 可以是 double 或 long long 等类型。在中间步骤中使用了乘积运算，因此如果使用 int 或 long long 类型，请注意溢出。
Usage: bool left = sideOf(p1,p2,q)==1;
"Point.h" 3af81c, 9 lines

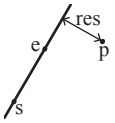
```
template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }

template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
  auto a = (e-s).cross(p-s); // 避免 s==e
  double l = (e-s).dist()*eps;
```

```
  return (a > l) - (a < -l);
}
```

lineDistance.h

Description:

返回点 p 到包含点 a 和 b 的直线的有符号距离。从 a 到 b 看，左侧为正值，右侧为负值。a==b 时返回 nan。P 应该是 Point<T> 或 Point3D<T> 类型，其中 T 可以是 double 或 long long 等类型。在中间步骤中使用了乘积运算，因此如果使用 int 或 long long 类型，请注意溢出。使用 Point3D 将始终给出非负距离。对于 Point3D，请在叉积的结果上调用.dist。



"Point.h" f6bf6b, 4 lines

```
template<class P>
double lineDist(const P& a, const P& b, const P& p) {
  return (double)(b-a).cross(p-a)/(b-a).dist();
}
```

求直线交点

$$1/0/-1 \Leftrightarrow \text{唯一交点/无交点/重合}$$

$$L_1 : s_1 + t(e_1 - s_1)$$

$$L_2 : s_2 + u(e_2 - s_2)$$

$$\text{交点} X = s_1 + t(e_1 - s_1) = s_2 + u(e_2 - s_2)$$

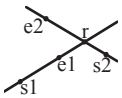
$$\begin{aligned}
\overrightarrow{s_2X} \parallel \overrightarrow{s_2e_2} &\Rightarrow (X - s_2) \times (e_2 - s_2) = 0 \\
&\Rightarrow (s_1 + t(e_1 - s_1) - s_2) \times (e_2 - s_2) = 0 \\
&\Rightarrow (s_1 - s_2) \times (e_2 - s_2) + t(e_1 - s_1) \times (e_2 - s_2) = 0 \\
&\Rightarrow t = \frac{(e_2 - s_2) \times (s_1 - s_2)}{(e_1 - s_1) \times (e_2 - s_2)}
\end{aligned}$$

$$\begin{aligned}
X =& s_1 + \frac{(e_2 - s_2) \times (s_1 - s_2)}{(e_1 - s_1) \times (e_2 - s_2)} \cdot (e_1 - s_1) \\
=& \{[(e_1 - s_1) \times (e_2 - s_2) - (e_2 - s_2) \times (s_1 - s_2)] \cdot s_1 \\
&+ [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1\} \cdot \frac{1}{(e_1 - s_1) \times (e_2 - s_2)} \\
=& \{[(e_1 - s_1) \times (e_2 - s_2) + (s_1 - s_2) \times (e_2 - s_2)] \cdot s_1 \\
&+ [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1\} \cdots \frac{1}{(e_1 - s_1) \times (e_2 - s_2)} \\
=& \frac{[(e_1 - s_2) \times (e_2 - s_2)] \cdot s_1 + [(e_2 - s_2) \times (s_1 - s_2)] \cdot e_1}{(e_1 - s_1) \times (e_2 - s_2)} \\
=& \frac{p \cdot s_1 + q \cdot e_1}{d}
\end{aligned}$$

lineIntersection.h

Description:

如果存在唯一的交点，返回 {1，交点}；如果不存在交点，返回 {0，(0,0)}；如果存在无数个交点，返回 {-1，(0,0)}。如果使用 `Point<ll>`，且交点坐标不全为整数，则返回错误的结果。中间步骤使用了三个坐标的乘积，因此如果使用 `int` 或 `ll`，请注意可能的溢出。



Usage: `auto res = lineInter(s1,e1,s2,e2);`
`if (res.first == 1)`
`cout << "intersection point at " << res.second << endl;`

"Point.h" a01f81, 8 lines

```
template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
  auto d = (e1 - s1).cross(e2 - s2);
  if (d == 0) // if parallel
    return {-(s1.cross(e1, s2) == 0), P(0, 0)};
  auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
  return {1, (s1 * p + e1 * q) / d};
}
```

Strings (5)

Various (6)

6.1 Misc. algorithms

LIS.h

Description: 返回最长上升（非降）子序列的下标。

Time: $\mathcal{O}(N \log N)$

277a90, 17 lines

```
template<class I> vi lis(const vector<I> &S) {
  if (S.empty()) return {};
  vi prev(sz(S));
  using p = pair<I, int>;
  vector<p> res;
  rep(i, 0, sz(S)) {
    // 把lower_bound中的 0 -> i 以求最长非降子序列
    auto it = lower_bound(all(res), p{S[i], 0});
    if (it == res.end()) res.emplace_back(), it = res.end()-1;
    *it = {S[i], i};
    prev[i] = it == res.begin() ? 0 : (it - 1)->second;
  }
  int L = sz(res), cur = res.back().second;
  vi ans(L);
  while (L--) ans[L] = cur, cur = prev[cur];
  return ans;
}
```